

ECEN 662:

Estimation & Detection Theory

Lecture 15: Minimax Estimators and Worst-Case Optimality (Part II)

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Estimating the mean of an unknown distribution

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$$X_i \stackrel{iid}{\sim} f(x_i)$$

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- To see this, let us first *restrict* f to be *Gaussian*. Under this restriction, previously we have shown that the sample mean $\bar{X}$ is minimax with a maximum risk B/n

- Further note that for any f , the risk of the sample mean $\bar{X}$ is

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- We may thus conclude that the sample mean $\bar{X}$ is minimax for all f with $\sigma_f^2 \leq B$
- What if the constraint is on the *support* of f instead, say the support of f is $[0, 1]$?

- Previously, we have shown that if f is *Bernoulli*, then the sample mean $\bar{X}$ is *not* minimax. Instead, a minimax estimator is given by

$$\delta(X) = \frac{\bar{X} + 1/(2\sqrt{n})}{1 + 1/\sqrt{n}}$$

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- To show that the above estimator is, in fact, minimax for *all* f supported in $[0, 1]$, all we need to do is to show that the maximum risk of δ remains to be

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- To see this, simply note that

$$\begin{aligned} R_\delta(f) &= \mathbb{E}_f[(\delta(X) - \theta_f)^2] \\ &= \frac{1}{(\sqrt{n} + 1)^2} \left(\mathbb{E}_f[X_1^2] - \theta_f + \frac{1}{4} \right) \end{aligned}$$

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- We now use the fact that $X_1 \in [0, 1]$ so we have $X_1^2 \leq X_1$ with probability 1 and hence

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- We may thus conclude that

$$R_\delta(f) \leq \frac{1}{4(\sqrt{n} + 1)^2}$$

for any f supported on $[0, 1]$, where the equality holds when f is *Bernoulli*

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- One situation for which the *uniqueness* of the minimax estimator can be readily recognized is as follows: If δ_π is a *unique* Bayes estimator such that

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- To see this, let δ' be another estimator that is *different* from δ_π . We have

$$R_{\delta'}^{(M)} \geq R_{\delta', \pi} > R_\pi^{(B)} = R_{\delta_\pi}^{(M)}$$

where the *strict* inequality follows from the *uniqueness* of δ_π as a Bayes estimator

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- To show that that $\bar{X}$ is admissible, assume the opposite so there exists another estimator δ that *strictly* dominates $\bar{X}$
- Note that the risk function of $\bar{X}$ is given by $R_{\bar{X}}(\theta) = \sigma^2/n$. By the fact that the risk function for any estimator is *continuous*, there must exist an interval (θ_0, θ_1) such that

$$R_{\delta}(\theta) \leq \sigma^2/n - \epsilon, \quad \forall \theta \in (\theta_0, \theta_1)$$

for some $\epsilon > 0$

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- The Bayesian risk of δ can be bounded from above as:

$$\begin{aligned}
R_{\delta, \pi_{\sigma_0^2}} &= \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{-\infty}^{\infty} R_{\delta}(\theta) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \\
&= \frac{1}{\sqrt{2\pi\sigma_0^2}} \left(\int_{-\infty}^{\theta_0} + \int_{\theta_0}^{\theta_1} + \int_{\theta_1}^{\infty} \right) R_{\delta}(\theta) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \\
&\leq \frac{1}{\sqrt{2\pi\sigma_0^2}} \left(\left(\int_{-\infty}^{\theta_0} + \int_{\theta_1}^{\infty} \right) \left(\frac{\sigma^2}{n} \right) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta + \right. \\
&\quad \left. \int_{\theta_0}^{\theta_1} \left(\frac{\sigma^2}{n} - \epsilon \right) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \right) \\
&= \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{-\infty}^{\infty} \left(\frac{\sigma^2}{n} \right) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta - \\
&\quad \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{\theta_0}^{\theta_1} \epsilon \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \\
&= \frac{\sigma^2}{n} - \frac{\epsilon}{\sqrt{2\pi\sigma_0^2}} \int_{\theta_0}^{\theta_1} \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta
\end{aligned}$$

- Note that when $\sigma_0^2 \rightarrow \infty$, both the upper bound of $R_{\delta, \pi_{\sigma_0^2}}$ and $R_{\delta_{\pi_{\sigma_0^2}}, \pi_{\sigma_0^2}}$ approach σ^2/n , but with *different* convergence rates

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- In particular, we have

$$\frac{\sigma^2/n - R_{\delta, \pi_{\sigma_0^2}}}{\sigma^2/n - R_{\delta_{\pi_{\sigma_0^2}}, \pi_{\sigma_0^2}}} \geq \frac{\frac{\epsilon}{\sqrt{2\pi\sigma_0^2}} \int_{\theta_0}^{\theta_1} \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta}{\frac{\sigma^4}{n(\sigma^2 + n\sigma_0^2)}}$$

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- Therefore, for sufficiently large σ_0^2 , we will have

$$\frac{\sigma^2/n - R_{\delta, \pi_{\sigma_0^2}}}{\sigma^2/n - R_{\delta_{\pi_{\sigma_0^2}}, \pi_{\sigma_0^2}}} > 1$$

and hence

$$R_{\delta, \pi_{\sigma_0^2}} < R_{\delta_{\pi_{\sigma_0^2}}, \pi_{\sigma_0^2}}$$

contradicting the fact that $\delta_{\pi_{\sigma_0^2}}$ is the *Bayes* estimator for $\pi_{\sigma_0^2}$

Randomized minimax estimators

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- Let $X \sim \text{Binomial}(n, \theta)$, where $\theta \in [0, 1]$ is the unknown parameter. Consider estimation of θ under the *0-1* loss:

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- Claim: If $\alpha < \frac{1}{2(n+2)}$, the maximum risk of *any* deterministic estimator is 1

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- Hence, in this setting, there can be *no* deterministic minimax estimator